

1 Data & runs

2 Curves

3 Model fit

4 Compare

5 Export

Data & runs

Pull every asset class, choose the segments, run. Models are fitted on all actuals up to the date you pick, then projected forward.

Actuals through

31 Aug 2026 (latest) ▾

1 Sources Each asset class runs its standard Hadoop query unless you swap in a file.

Asset class	Source		What will run	
Investments	Hadoop query	CSV file	Standard query 612K rows on last pull, 14 Sep, 07:11	Adjust
Deposits	Hadoop query	CSV file	Standard query with your adjustments 1 change to the standard query	Showing
Mortgage	Hadoop query	CSV file	Standard query 288K rows on last pull, 14 Sep, 07:12	Adjust
Credit cards	Hadoop query	CSV file	Standard query 1.6M rows on last pull, 14 Sep, 07:13	Adjust

2 Segments

- ☒ Customer typeCross-sell, Prospect
- ☐ ChannelBranch, Digital, Phone
- ☐ Region5 regions
- ☐ ProductVaries by class

+ New segmentation

Group any column’s values into a new cut, such as one finance asks for. It’s saved and available on every run after.

Deposits — Hadoop query

Reset to standard

The standard query is maintained by analytics and never changes. Adjustments here are saved as a copy for this asset class.

Filters

Opened on or after 1 Sep 2023

Actuals through the selected date

Region is not TEST

Also break out by

Nothing extra

Adds a column you can then pick as a segment.

Last pulled 14 Sep, 07:12, 2.9M rows

Query that will run

[Edit SQL directly](#)

```
1 SELECT
2   cohort_month,
3   months_since_open      AS period,
4   customer_type,
5   COUNT(*)               AS active_custc
6 FROM retail.deposits_account_month
7 WHERE open_date >= '2023-09-01'
8    AND snapshot_month <= '2026-08-31'
9    AND is_active = 1
10   AND region <> 'TEST'
11 GROUP BY cohort_month, months_since_open,
12           customer_type
```

Highlighted lines come from your adjustments.

3 Model choice

- ☒ Likelihood-ratio test (recommended)
Use BdW where it fits significantly better than sBG at the 5% level; otherwise use sBG.
- ☐ Always BdW
Simplest to explain; BdW includes sBG as a special case.
- ☐ Always sBG
Only if you need to match older curves.

Project forward to

60 months

120 months

Four asset classes, each split by customer type into 2 segments, plus a total per class: 12 curves. Each is fitted with both models and projected to month 60.

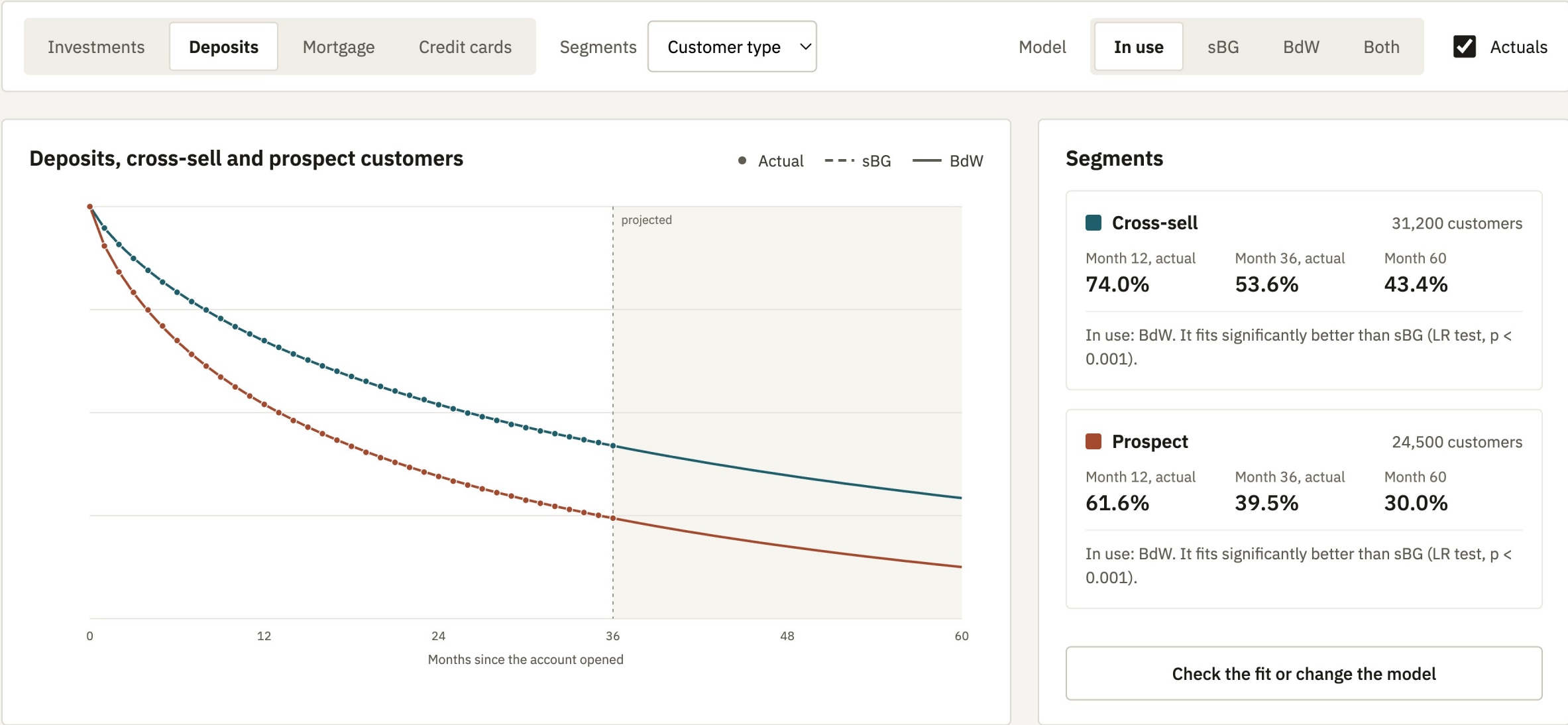
- ☐ Pull datawaiting
- ☐ Check columns and dateswaiting
- ☐ Build cohort tableswaiting
- ☐ Fit sBG and BdWwaiting
- ☐ Choose model per segmentwaiting

Run all asset classes

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Curves

Share of each starting group still active, month by month. Dots are actuals; lines are the fitted models, projected past the last actual month.



By cohort — Deposits, prospect customers

Each row is the customers who opened in that month. Read down a column to see whether newer cohorts are holding on better or worse.

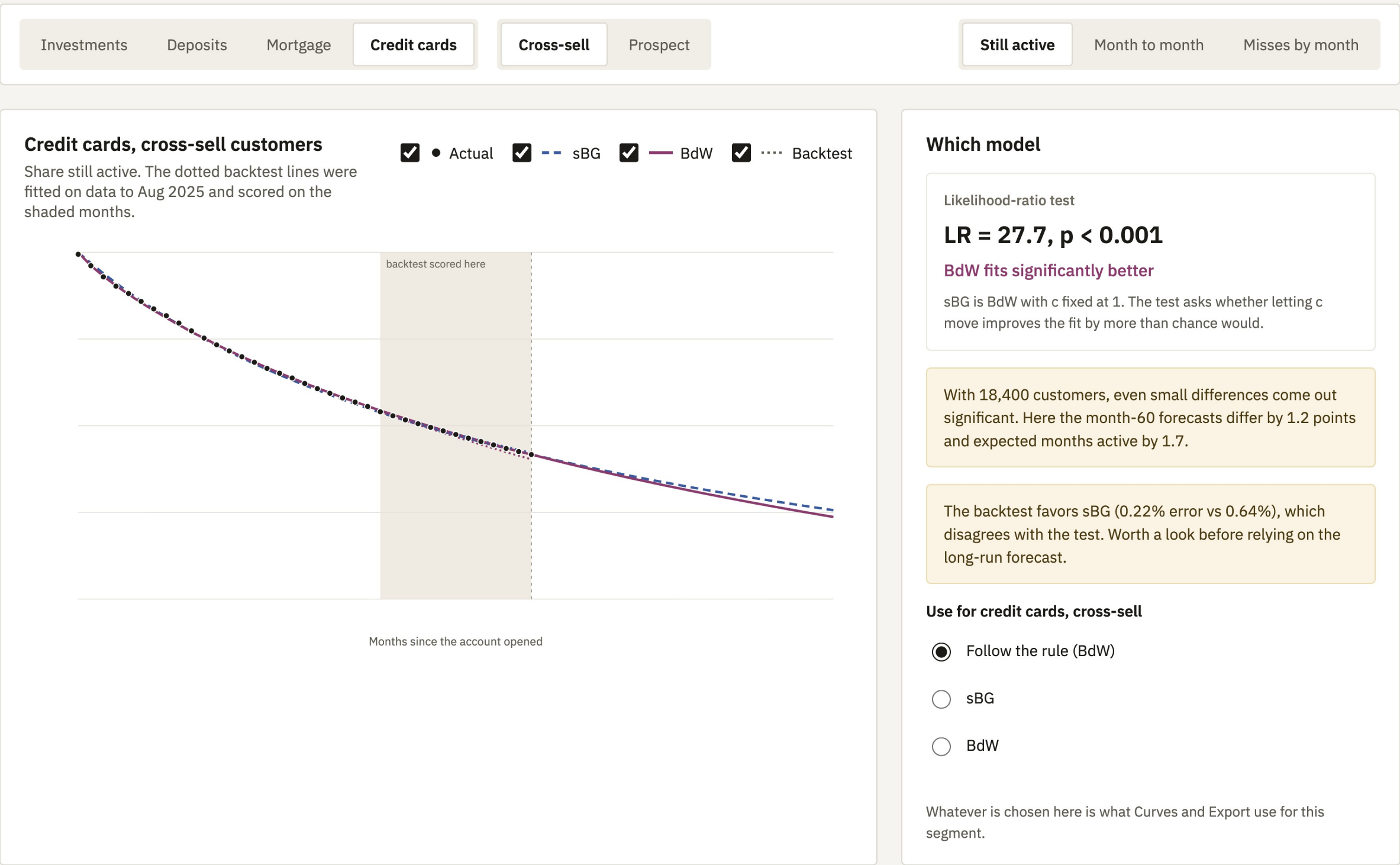
Cohort	Customers	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
Aug 2024	743	93	87	82	79	76	73	71	68	66	64	63	61	59	57	56	55	54	52	51	51	50	48	47	47
Sep 2024	803	92	87	83	80	76	73	70	69	67	65	62	61	60	58	57	55	54	53	52	51	49	49	48	
Oct 2024	729	92	88	83	79	76	74	72	69	66	64	63	62	60	58	57	56	55	53	52	51	50	49		
Nov 2024	602	93	87	83	80	77	74	71	69	67	66	64	62	60	59	58	56	55	53	53	52	51			
Dec 2024	561	92	87	84	81	77	74	72	70	68	65	63	62	61	59	58	56	55	54	53	52				
Jan 2025	650	93	88	84	80	77	75	72	70	67	66	64	63	61	59	58	57	56	54	53					
Feb 2025	773	93	87	83	80	78	75	72	70	68	66	64	62	61	60	58	57	55	54						
Mar 2025	794	92	88	84	81	77	74	72	70	68	66	64	63	61	60	58	57	56							
Apr 2025	692	93	88	83	80	77	75	72	70	67	66	64	63	61	59	58	57								
May 2025	577	92	87	83	81	77	74	71	70	68	66	64	62	61	60	58									
Jun 2025	575	92	88	84	80	76	74	72	70	67	65	64	62	61	59										
Jul 2025	689	93	88	83	80	77	74	72	69	67	66	64	62	60											
Aug 2025	793	92	87	83	80	77	74	71	69	68	65	63	62												
Sep 2025	775	92	88	84	80	76	74	72	69	67	65	64													
Oct 2025	653	93	87	83	80	77	74	71	69	67	66														
Nov 2025	562	92	87	84	81	77	74	72	70	68															
Dec 2025	600	93	88	84	80	77	75	73	70																
Jan 2026	726	93	88	84	81	78	75	72																	
Feb 2026	803	92	88	85	81	78	75																		
Mar 2026	745	94	89	84	81	78																			
Apr 2026	617	93	88	84	82																				
May 2026	558	93	89	85																					
Jun 2026	633	94	88																						
Jul 2026	760	92																							

40%100% still active

Column headers are months since opening. Blank cells are months a cohort hasn't reached.

Model fit

Both models are fitted on every month of actuals. The likelihood-ratio test picks between them; the backtest checks how each would have forecast the last year.



Fit and forecast checks

Measure	sBG	BdW	What it tells you
Fit to the actuals, months 1–36			
Log-likelihood	-34,527.8	-34,513.9	Higher is better. The LR test is twice the gap between these two.
AIC	69,059.5	69,033.8	Lower is better. Small charge for BdW's extra parameter.
BIC	69,075.2	69,057.3	Lower is better. Bigger charge that grows with customer count, so it resists over-fitting large segments.
Average miss	0.23%	0.14%	Mean absolute percentage gap between the curve and the actuals.
Largest miss	0.61 pts	0.37 pts	The worst single month, in percentage points.
Forecast test			
Backtest error	0.22%	0.64%	Refitted on data to Aug 2025, then scored on the 12 months since. The only check here on data the model hadn't seen.
Parameters, with 95% range			
α	0.49 (0.44–0.54)	1.05 (0.66–1.67)	With β , sets how churn risk is spread across customers (chart below).
β	25.6 (22.1–29.5)	44.9 (29.5–68.4)	Wide ranges mean the data can't pin the model down; expect the forecast to move between runs.
c	1 (fixed)	0.87 (0.83–0.92)	Below 1: a customer's churn chance falls the longer they stay. Above 1: it rises.
What marketing will use			
Still active at month 60	55.4%	54.2%	The long-run end of the curve, where the two models can disagree most.
Expected months active, first 10 years	71.4	69.7	Sum of the curve to month 120. This is the lifetime figure an LTV is built on.
Median lifetime	81 months	73 months	The month by which half of the customers have left.
Month-60 change since last run	–0.6 pts	–0.5 pts	Against the August run. Large swings month to month deserve an explanation before they reach an LTV model.

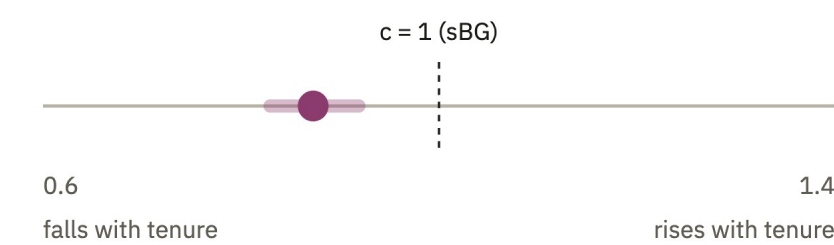
How each model spreads churn risk across customers

sBG explains the shape of the curve only through differences between customers, so it has to assume a particular spread of risk (average first-month churn chance 1.9%). BdW can also let each customer's own risk change with tenure, so its spread can differ (average 2.3%). When the two shapes look alike, c is close to 1.



Does churn change with tenure?

BdW's shape value c answers this for a single customer, separately from the mix of customers. If its range crosses 1, the data can't tell, and sBG is enough.



The range (0.83–0.92) sits below 1: a customer who has stayed a while becomes less likely to leave, on top of risky customers leaving early.

Every segment in this run

Flags mark segments worth a second look before marketing uses them.

Asset class	Segment	Customers	LR test	Backtest, sBG	Backtest, BdW	Month 60 vs last run	Using	
Investments	Cross-sell	14,800	= 0.80	0.21%	0.33%	+0.0 pts	sBG, by the rule	View
Investments	Prospect	9,600	< 0.001	1.27%	0.38%	+0.1 pts	BdW, by the rule	View
Deposits	Cross-sell	31,200	< 0.001	0.98%	0.13%	−0.3 pts	BdW, by the rule	View

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Compare

Follow one segment across time: customers who opened in different periods, or the curve as it looked with older data.

Segments

Time periods

Segment

Investments, cross-sell

Compare periods by

When customers opened

Data as of

Period length

Month

Year

Show

Actuals only

Actuals and refitted curves

Opened Jul 2026

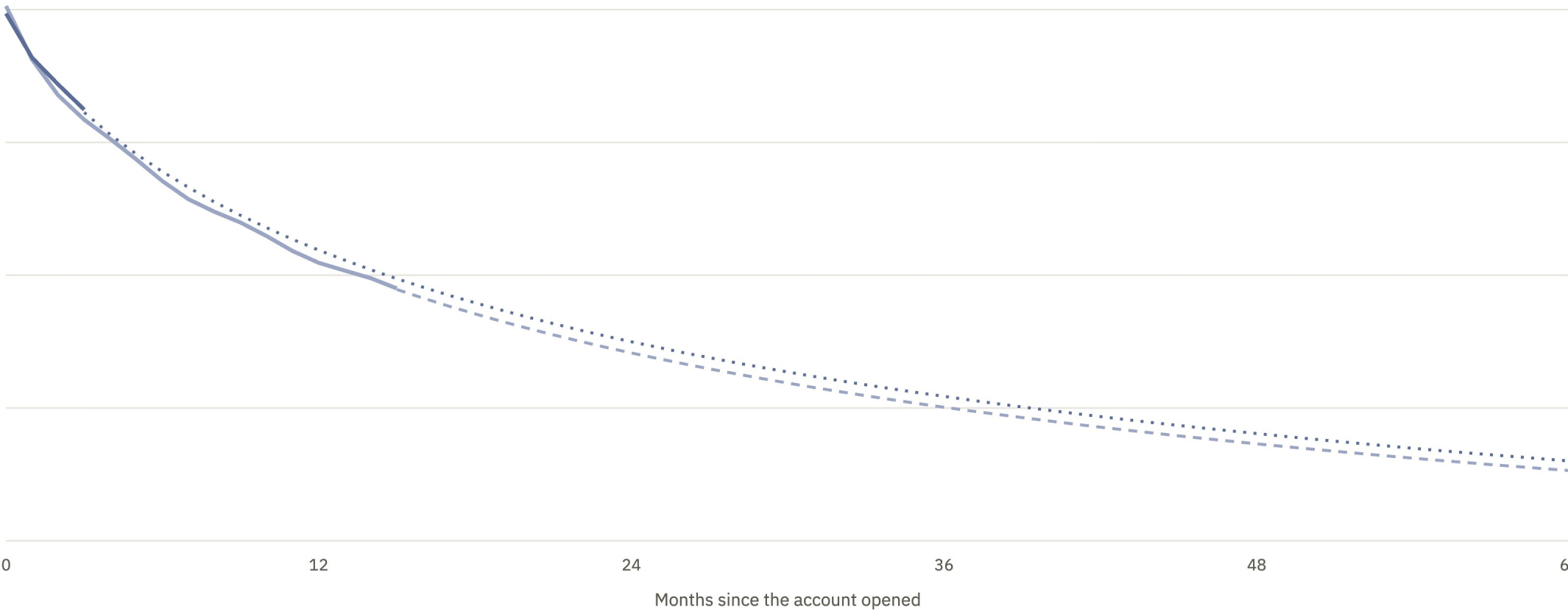
Opened Jul 2026

+ Add period

Each period refitted on its own actuals.

Deposits, prospect customers

Solid: actual. Dashed: refitted curve. Dotted: refit on too few months to trust.



Side by side

Share still active, actuals. The first period is the baseline.

	Customers	Months of actuals	Month 3	Month 6	Month 12	Month 60, refit	Forecast
Opened May 2025	577	15	83.4% baseline	74.2% baseline	61.9% baseline	30.6%	Rough treat as indicative
Opened May 2026	558	3	84.9% +1.5 pts	— not yet	— not yet	32.0%	Too early shape is mostly assumed

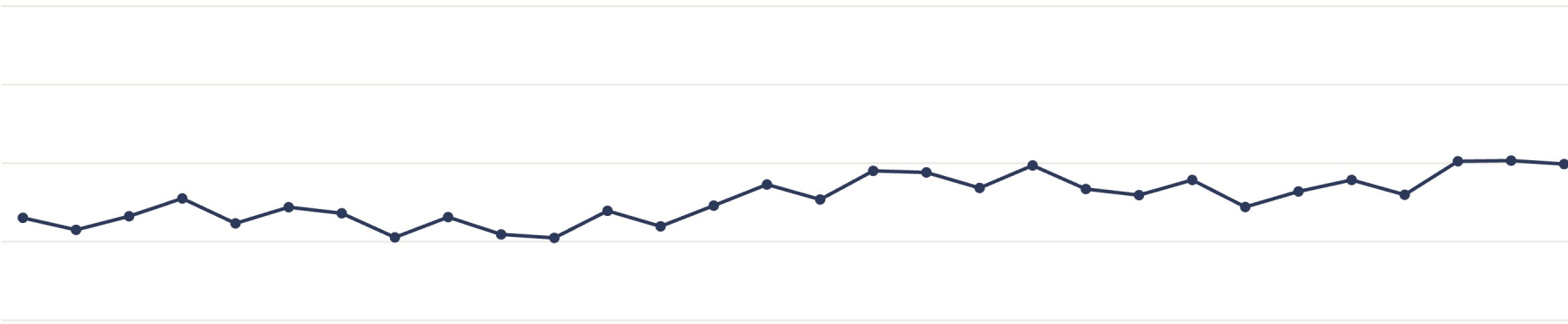
Still active at month 6, by the month customers opened

Actuals only, one point per monthly cohort. The quickest way to spot when retention shifted.

Month 3

Month 6

Month 12



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Export

Pick the curves and a format. Copy is quickest for pasting into a model; CSV and Excel are for keeping; the image is for slides.

Curves

8 curves

☐ Investments

2 of 3

☐ All customers

In use (BdW)

▼

☒ Cross-sell

In use (sBG)

▼

☒ Prospect

In use (BdW)

▼

☐ Deposits

2 of 3

☐ All customers

In use (BdW)

▼

☒ Cross-sell

In use (BdW)

▼

☒ Prospect

In use (BdW)

▼

☐ Mortgage

2 of 3

☐ All customers

In use (BdW)

▼

☒ Cross-sell

In use (BdW)

▼

☒ Prospect

In use (BdW)

▼

☐ Credit cards

2 of 3

☐ All customers

In use (BdW)

▼

☒ Cross-sell

In use (BdW)

▼

☒ Prospect

In use (BdW)

▼

“All customers” blends the two segment curves by customer count.

Copy for pasting

Tab-separated, straight into a spreadsheet or LTV model.

CSV

One file you can keep, load or share.

Excel workbook

A sheet per asset class, plus heat maps and a README.

Chart image

PNG of the selected curves, sized for a slide.

Months

Values as

24

60

120

Percent

Decimal

☐ Actuals next to each curve

☒ Customer counts

Preview

1 more columns not shown. Months down, one column per curve

	A	B	C	D	E	F	G	H
1	Month	Investme...	Investme...	Deposits ...	Deposits ...	Mortgage ...	Mortgage ...	Credit car...
2	Customers	14,800	9,600	31,200	24,500	6,100	4,300	18,400
3	0	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%
4	1	98.6%	96.9%	95.8%	92.4%	99.7%	99.4%	97.7%
5	2	97.2%	94.6%	92.7%	87.4%	99.3%	98.7%	95.9%
6	3	95.9%	92.6%	90.0%	83.2%	98.9%	98.0%	94.3%
7	4	94.7%	90.9%	87.6%	79.7%	98.4%	97.3%	92.7%
8	5	93.5%	89.2%	85.4%	76.6%	97.9%	96.5%	91.3%
9	6	92.3%	87.7%	83.4%	73.9%	97.4%	95.7%	90.0%
10	7	91.2%	86.2%	81.6%	71.4%	96.8%	95.0%	88.7%
11	8	90.1%	84.9%	79.8%	69.1%	96.3%	94.2%	87.4%
12	9	89.0%	83.6%	78.2%	67.0%	95.7%	93.4%	86.3%
13	10	88.0%	82.3%	76.7%	65.1%	95.1%	92.6%	85.1%
14	11	87.0%	81.1%	75.3%	63.3%	94.5%	91.8%	84.0%
15	...							
16	60	59.0%	50.6%	43.4%	30.0%	68.5%	57.2%	54.2%

Clipboard

Copy 8 curves

Paste into cell A1 of your sheet.